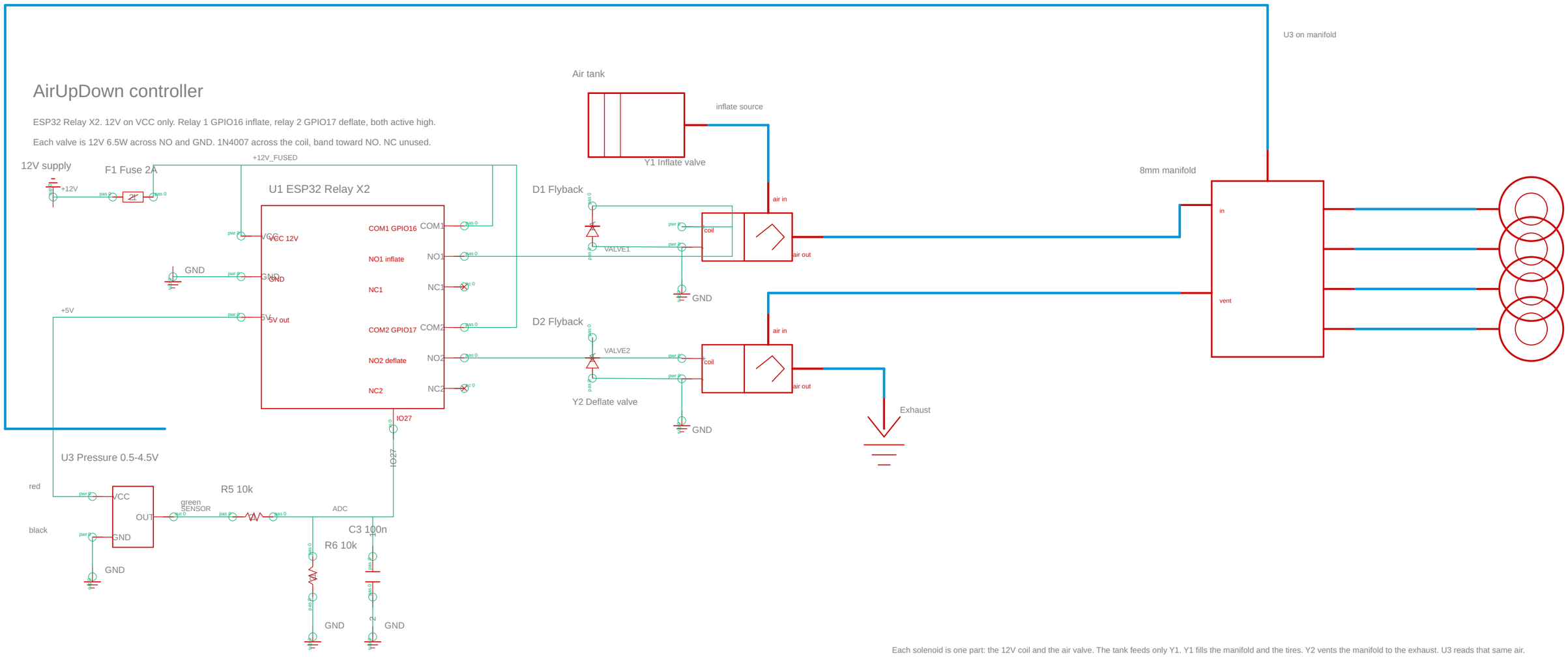
AirUpDown controller

ESP32 Relay X2. 12V on VCC only. Relay 1 GPIO16 inflate, relay 2 GPIO17 deflate, both active high.

Each valve is 12V 6.5W across NO and GND. 1N4007 across the coil, band toward NO. NC unused.



U3 red +5V, black GND, green signal 0.5-4.5V. 10k/10k divider into GPIO27. Full scale is the PSI stamped on the sensor.

Each solenoid is one part: the 12V coil and the air valve. The tank feeds only Y1. Y1 fills the manifold and the tires. Y2 vents the manifold to the exhaust. U3 reads that same air.